for any sequence of data and nonnegative losses. Hint: replace Hoeffding's Lemma by (1.40).

3. Suppose $\ell(f(x), y) \in [0, 1]$ for all $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $f \in \mathcal{F}$. Suppose that there is a "perfect expert $f^\star \in \mathcal{F}$ such that $\ell(f^\star(x^t), y^t) = 0$ for all $t \in [T]$. Conclude that the above algorithm, with an appropriate choice of η, enjoys a bound of $O(\log|\mathcal{F}|)$ on the cumulative loss of the algorithm (equivalently, the fast rate $\frac{\log|\mathcal{F}|}{T}$ for the average regret). This setting is called "zero-noise."

4. Consider the binary classification problem with indicator loss, and suppose $\mathcal{F}$ contains a perfect expert, as above. The *Halving Algorithm* maintains a version space $\mathcal{F}^t = \{f \in \mathcal{F} : f(x^s) = y^s, s < t\}$ and, given x^t, follows the majority vote of remaining experts in $\mathcal{F}^t$. Show that this algorithm incurs cumulative loss at most $O(\log|\mathcal{F}|)$. Hence, the Exponential Weights Algorithm can be viewed as an extension of the Halving algorithm to settings where the optimal loss is non-zero.

2. MULTI-ARMED BANDITS

This chapter introduces the *multi-armed bandit* problem, which is the simplest interactive decision making framework we will consider in this course.

Multi-Armed Bandit Protocol

for $t = 1, \ldots, T$ **do** T rounds

  Select decision $\pi^t \in \Pi := \{1, \ldots, A\}$. (Select a fixed, discrete decision (=action) π^t, using the 지금까지 모아온 data!)

  Observe reward $r^t \in \mathbb{R}$.

The protocol (see above) proceeds in T rounds. At each round $t \in [T]$, the learning agent selects a discrete *decision*[2] $\pi^t \in \Pi = \{1, \ldots, A\}$ using the data

$$\mathcal{H}^{t-1} = \{(\pi^1, r^1), \ldots, (\pi^{t-1}, r^{t-1})\}$$

collected so far; we refer to Π as the *decision space* or action space, with $A \in \mathbb{N}$ denoting the size of the space. We allow the learner to randomize the decision at step t according to a distribution $p^t = p^t(\cdot \mid \mathcal{H}^{t-1})$, sampling $\pi^t \sim p^t$. Based on the decision π^t, the learner receives a reward r^t, and their goal is to maximize the cumulative reward across all T rounds. As an example, one might consider an application in which the learner is a doctor (or personalized medical assistant) who aims to select a treatment (the decision) in order to make a patient feel better (maximize reward); see Figure 4.

The multi-armed bandit problem can be studied in a stochastic framework, in which rewards are generated from a fixed (conditional) distribution, or an non-stochastic/adversarial framework in the vein of online learning (Section 1.6). We will focus on the stochastic framework, and make the following assumption.

Assumption 1 (Stochastic Rewards): Rewards are generated independently via

rewards are generated from

$$r^t \sim M^\star(\cdot \mid \pi^t), \tag{2.1}$$

a "fixed" (즉, conditional on π^t) dist.

where $M^\star(\cdot \mid \cdot)$ is the underlying *model* (conditional distribution).

[2]In the literature on bandits, decisions are often referred to as *actions*. We will use these terms interchangeably throughout this section.

We define

mean reward

$$f^\star(\pi) := \mathbb{E}[r \mid \pi] \tag{2.2}$$

as the mean reward function under $r \sim M^\star(\cdot \mid \pi)$. We measure the learner's performance via regret to the action $\pi^\star := \arg\max_{\pi \in \Pi} f^\star(\pi)$ with highest reward:

*In Ch1. ("loss" instead of reward)

$$\mathbf{Reg} = \sum_{t=1}^{T} \ell(\widehat{f}^t(x^t), y^t) - \min_{f \in \mathcal{F}} \sum_{t=1}^{T} \ell(f(x^t), y^t)$$

$\hat{f}, \mathcal{F}$ emp. minimizer $\hat{f}$ hyp. class

$$\mathbf{Reg} = \sum_{i=1}^{T} \mathbb{E}_{\widehat{f}^t \sim q^t}[\ell(\widehat{f}^t(x^t), y^t)] - \min_{f \in \mathcal{F}} \sum_{i=1}^{T} \ell(f(x^t), y^t).$$

를 뒤집어서

→대신 highest reward로...

$$\mathbf{Reg} := \sum_{t=1}^{T} f^\star(\pi^\star) - \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[f^\star(\pi^t)\right]. \tag{2.3}$$

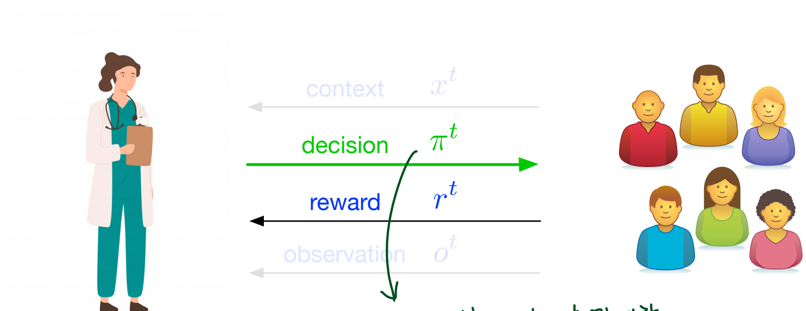
mean reward

Regret is a natural notion of performance for the multi-armed bandit problem because it is *cumulative:* it measures not just how well the learner can identify an action with good reward, but how well it can maximize reward as it goes. This notion is well-suited to settings like the personalized medicine example in Figure 4, where regret captures the *overall* quality of treatments, not just the quality of the final treatment. As in the online learning framework, we would like to develop algorithms that enjoy sublinear regret, i.e.

cumulative across all T rounds.

$$\frac{\mathbb{E}[\mathbf{Reg}]}{T} \to 0 \quad \text{as} \quad T \to \infty.$$

The most important feature of the multi-armed bandit problem, and what makes the problem fundamentally *interactive*, is that the learner only receives a reward signal for the single decision $\pi^t \in \Pi$ they select at each round. That is, the observed reward r^t gives a noisy estimate for $f^\star(\pi^t)$, but reveals no information about the rewards for other decisions $\pi \neq \pi^t$. For example in Figure 4, if the doctor prescribes a particular treatment to the



other possible treatment 에 대한 정보는 X.

Figure 4: An illustration of the multi-armed bandit problem. A doctor (the learner) aims to select a treatment (the decision) to improve a patient's vital signs (the reward).

patient, they can observe whether the patient responds favorably, but they do not directly observe whether other possible treatments might have led to an even better outcome. This issue is often referred to as *partial feedback* or *bandit feedback*. Partial feedback introduces an element of *active data collection*, as it means that the information contained in the dataset $\mathcal{H}^t$ depends on the decisions made by the learner, which we will see necessitates *exploring* different actions. This should be contrasted with statistical learning (where the dataset is generated independently from the learner) and online learning (where losses may be chosen by nature in response to the learner's behavior, but where the outcome y^t— and hence the full loss function $\ell(\cdot, y^t)$—is always revealed).

In the context of Figure 2, the multi-armed bandit problem constitutes our first step along the "interactivity" axis, but does not incorporate any structure in the decision space (and does not involve features/contexts/covariates). In particular, information about one action does not reveal information about any other actions, so there is no hope of using function approximation to generalize across actions.[3] As a result, the algorithms we will cover in this section will have regret that scales with $\Omega(|\Pi|) = \Omega(A)$. This shortcoming is addressed by the *structured bandit* framework we will introduce in Section 4, which allows for the use of function approximation to model structure in the decision space.[4]

A: size of the decision space.

Remark 8 (Other notions of regret): It is also reasonable to consider empirical regret, defined as

$$\max_{\pi \in \Pi} \sum_{t=1}^{T} r^t(\pi) - \sum_{t=1}^{T} r^t(\pi^t), \tag{2.4}$$

where, for $\pi \neq \pi^t$, $r^t(\pi)$ denotes the *counterfactual reward* the learner would have received if they "had played" π at round t. Using Hoeffding's inequality, one can show that this is equivalent to the definition in (2.3) up to $O(\sqrt{T})$ factors.

2.1 The Need for Exploration

In statistical learning, we saw that the empirical risk minimization algorithm, which greedily chooses the function that best fits the data, leads to interesting bounds on excess risk. For multi-armed bandits, since we assume the data generating process is stochastic, a natural first attempt at designing an algorithm is to apply the greedy principle here in the same fashion. Concretely, at time t, we can compute an empirical estimate for the reward function $f^\star$ via

emperical estimate...

$$\widehat{f}^t(\pi) = \frac{1}{\underbrace{n^t(\pi)}_{\text{얼마나 자주 }\pi\text{가 선택되었는지.}}} \sum_{s<t} r^s \mathbb{I}\{\pi^s = \overset{\text{기준!}}{\pi}\}, \tag{2.5}$$

...for the reward func f^* $(= \mathbb{E}[r|\pi])$

where $n^t(\pi)$ is the number of times π has been selected up to time t.[5] Then, we can choose the greedy action

the greedy action

$$\widehat{\pi}^t = \arg\max_{\pi \in \Pi} \widehat{f}^t(\pi).$$

Unfortunately, due to the interactive nature of the bandit problem, this strategy can fail, leading to linear regret ($\mathbf{Reg} = \Omega(T)$). Consider the following problem with $\Pi = \{1, 2\}$ ($A = 2$).

decision 선택지: 2개

- Decision 1 has reward $\frac{1}{2}$ almost surely.
- Decision 2 has reward $\mathrm{Ber}(3/4)$.

[3] Another way to say this is that we take $\mathcal{F} = \mathbb{R}^A$, so that $f^\star \in \mathcal{F}$.

[4] Throughout the lecture notes, we will exclusively use the term "multi-armed bandit" to refer to bandit problems with finite action spaces, and use the term "structured bandit" for problems with large action spaces.

[5] If $n^t(\pi) = 0$, we will set $\widehat{f}^t(\pi) = 0$.

Action 1: reward 1/2
Action 2: 〃 Ber(3/4)

첫 2번의 play 중, Action 2가 $\frac{1}{2}$ 미만의 보상을 받을 확률: $\frac{1}{4}$
↳이 경우, $\hat{f}_2 < \hat{f}_1 = \frac{1}{2}$ → 최적 action 인 Action 2를 더이상 방문 X.
regret $(f_2 - f_1 = \frac{1}{4})$ 증가 $(Reg_T = T(f_2 - f_1))$.

Suppose we initialize by playing each decision a single time to ensure that $n^t(\pi) > 0$, then follow the greedy strategy. One can see that with probability 1/4, the greedy algorithm will get stuck on action 1, leading to regret $\Omega(T)$.

↳ The case where Action 2 yields less than $\frac{1}{2}$ during the first two trials.

The issue in this example is that the greedy algorithm immediately gives up on the optimal action and never revisits it. To address this, we will consider algorithms that deliberately *explore* "less visited actions" to ensure that their estimated rewards are not misleading.

2.2 The ε-Greedy Algorithm

alg. that deliberately explore LESS visited actions

The greedy algorithm for bandits can fail because it can insufficiently explore good decisions that initially seem bad, leading it to get stuck playing suboptimal decisions. In light of this failure, a reasonable solution is to manually force the algorithm to explore, so as to ensure that this situation never occurs. This leads us to what is known as the *ε-Greedy* algorithm (e.g., Sutton and Barto [81], Auer et al. [13]).

Let $\varepsilon \in [0, 1]$ be the *exploration parameter*. At each time $t \in [T]$, the ε-Greedy algorithm computes the estimated reward function $\widehat{f}^t$ as in (2.5). With probability $1 - \varepsilon$, the algorithm chooses the greedy decision

" $\hat{f}^t(\pi) = \frac{1}{n^t(\pi)} \sum_{s<t} r^s \mathbb{1}\{\pi^s = \pi\}$ "

the greedy action $\hat{\pi}^t$:
$$\widehat{\pi}^t = \arg\max_{\pi} \widehat{f}^t(\pi), \tag{2.6}$$

and with probability ε it samples a uniform random action $\pi^t \sim \mathrm{unif}(\{1, \ldots, A\})$. As the name suggests, ε-Greedy usually plays the greedy action (*exploiting* what it has already learned), but the uniform sampling ensures that the algorithm will also *explore* unseen actions. We can think of the parameter ε as modulating the tradeoff between exploiting and exploring.

mean reward func
$\mathbb{E}[r|\pi]$

Proposition 4: Assume that $f^\star(\pi) \in [0, 1]$ and r^t is 1-sub-Gaussian. Then for any T, by choosing ε appropriately, the ε-Greedy algorithm ensures that with probability at least $1 - \delta$,

$\Pr\left(\mathbb{E}[\mathrm{Reg}] \gtrsim A^{1/3}T^{2/3} \cdot \log^{1/3}(AT/\delta)\right) \leq \delta.$

$$\mathbb{E}[\mathbf{Reg}] \lesssim A^{1/3}T^{2/3} \cdot \log^{1/3}(AT/\delta).$$

$Reg \leq \sqrt{\frac{AT \log(AT/\delta)}{\varepsilon}} + \varepsilon T$ where $\varepsilon \propto \left(\frac{A \log(AT/\delta)}{T}\right)^{1/3}$

This regret bound has $\frac{\mathbb{E}[\mathbf{Reg}]}{T} \to 0$ with $T \to \infty$ as desired, though we will see in the sequel that more sophisticated strategies can attain improved regret bounds that scale with $\sqrt{AT}$.[6]

ε-Greedy alg. allows the learner to acquire information uniformly from all actions
BUT...
we PAY for this in "regret"
→ the εT factor

Proof of Proposition 4. Recall that $\widehat{\pi}^t := \arg\max_\pi \widehat{f}^t(\pi)$ denotes the greedy action at round t, and that p^t denotes the distribution over π^t. We can decompose the regret into two terms, representing the contribution from choosing the greedy action and the contribution from

the greedy action
↳ exploiting

[6] Note that $\sqrt{AT} \leq A^{1/3}T^{2/3}$ whenever $A \leq T$, and when $A \geq T$ both guarantees are vacuous.

exploring uniformly:

explore

exploit (choose the greedy action $\widehat{\pi}^t$)

$$
\begin{aligned}
\mathbf{Reg} &= \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(\pi^\star) - f^\star(\pi^t)] \\
&= (1-\varepsilon) \sum_{t=1}^{T} \left(f^\star(\pi^\star) - f^\star(\widehat{\pi}^t)\right) + \varepsilon \underbrace{\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim \mathrm{unif}([A])}[f^\star(\pi^\star) - f^\star(\widehat{\pi}^t)]}_{\le \sum_{t=1}^{T} 1 = T.} \\
&\le \sum_{t=1}^{T} \left(f^\star(\pi^\star) - f^\star(\widehat{\pi}^t)\right) + \varepsilon T.
\end{aligned}
$$

In the last inequality, we have simply written off the contribution from exploring uniformly by using that $f^\star(\pi) \in [0,1]$. It remains to bound the regret we incur from playing the greedy action. Here, we bound the per-step regret in terms of estimation error using a similar decomposition to Lemma 4 (note that we are now working with "rewards" rather than losses):

"reward" 비교

$$
\left(f^\star(\pi^\star) - f^\star(\widehat{\pi}^t)\right) = [f^\star(\pi^\star) - \widehat{f}^t(\pi^\star)] + \underbrace{[\widehat{f}^t(\pi^\star) - \widehat{f}^t(\widehat{\pi}^t)]}_{\le 0} + [\widehat{f}^t(\widehat{\pi}^t) - f^\star(\widehat{\pi}^t)] \tag{2.7}
$$

$$
\le 2 \max_{\pi \in \{\pi^\star, \widehat{\pi}^t\}} |f^\star(\pi) - \widehat{f}^t(\pi)| \le 2 \max_{\pi} |f^\star(\pi) - \widehat{f}^t(\pi)|. \tag{2.8}
$$

Note that this regret decomposition can also be applied to the pure greedy algorithm, which we have already shown can fail. The reason why ε-Greedy succeeds, which we use in the argument that follows, is that because we explore, the "effective" number of times that each arm will be pulled prior to round t is of the order $\varepsilon t/A$, which will ensure that the sample mean converges to $f^\star$. In particular, we will show that the event

$$
\mathcal{E}_t = \left\{ \max_{\pi} |f^\star(\pi) - \widehat{f}^t(\pi)| \lesssim \sqrt{\frac{A \log(AT/\delta)}{\varepsilon t}} \right\} \tag{2.9}
$$

occurs for all t with probability at least $1-\delta$.

To prove that (2.9) holds, we first use Hoeffding's inequality for adaptive stopping times (Lemma 33), which gives that for any fixed π, with probability at least $1-\delta$ over the draw of rewards,

Hoeffding : $\text{emp mean} - \text{mean} \le \overset{구간길이}{(b-a)} \sqrt{\frac{\log(T/\delta)}{2T'}}$ $\forall T'$ s.t. $1 \le T' \le T$. $[1, 2T]$

$|\text{emp mean} - \text{mean}| \le \cancel{2} \times (1-0) \sqrt{\frac{2\log(2T/\delta)}{\cancel{2}\, n^t(\pi)}}$

$$
|\underbrace{f^\star(\pi)}_{\text{mean}} - \underbrace{\widehat{f}^t(\pi)}_{\text{emp.mean}}| \le \sqrt{\frac{2\log(2T/\delta)}{n^t(\pi)}}. \tag{2.10}
$$

From here, taking a union bound over all $t \in [T]$ and $\pi \in \Pi$ (size: A) ensures that

$[1, 2AT^2]$

$$
|f^\star(\pi) - \widehat{f}^t(\pi)| \le \sqrt{\frac{2\log(2AT^2/\delta)}{n^t(\pi)}} \tag{2.11}
$$

for all π and t simultaneously. It remains to show that the number of pulls $n^t(\pi)$ is sufficiently large.

Let $e^t \in \{0,1\}$ be a random variable whose value indicates whether the algorithm explored uniformly at step t, and let $m^t(\pi) = |\{i < t : \pi^i = \pi, e^i = 1\}|$, which has $n^t(\pi) \ge m^t(\pi)$. Let $Z^t = \mathbb{I}\{\pi^t = \pi, e^t = 1\}$. Observe that we can write

how many times π was selected up to time t

"uniformly explore" 한 횟수 up to (before) time t.

$$
m^t(\pi) = \sum_{i<t} Z^i.
$$

걍 같은 뜻.

$m^t(\pi) = \sum_{i<t} Z^i$

In addition, $Z^t \sim \text{Ber}(\varepsilon/A)$, so we have $\mathbb{E}[m^t(\pi)] = \varepsilon(t-1)/A$. Using Bernstein's inequality (Lemma 5) with $Z^1, \ldots, Z^{t-1}$, we have that for any fixed π and all $u > 0$, with probability at least $1 - 2e^{-u}$,

$|m^t(\pi) - \mathbb{E}[m^t(\pi)]| \leq \sqrt{2 \frac{\varepsilon}{A} \cdot \underbrace{(1 - \frac{\varepsilon}{A})}_{\leq 1} (t-1)u} + \frac{u}{3} \leq \sqrt{2 \cdot \frac{\varepsilon}{A}(t-1)u} + \frac{u}{3} \underset{\text{산술기하}}{\leq} \frac{2u}{2} + \frac{\varepsilon(t-1)}{2A} + \frac{u}{3} = \frac{\varepsilon(t-1)}{2A} + \frac{4u}{3}$

$$\left| m^t(\pi) - \frac{\varepsilon(t-1)}{A} \right| \leq \sqrt{2\mathbb{V}[Z](t-1)u} + \frac{u}{3} \leq \sqrt{\frac{2\varepsilon(t-1)u}{A}} + \frac{u}{3} \leq \frac{\varepsilon(t-1)}{2A} + \frac{4u}{3},$$

where we have used that $\mathbb{V}[Z] = \varepsilon/A \cdot (1 - \varepsilon/A) \leq \varepsilon/A$, and then applied the arithmetic mean-geometric mean (AM-GM) inequality, which states that $\sqrt{xy} \leq \frac{x}{2} + \frac{y}{2}$ for $x, y \geq 0$. Rearranging, this gives

$-\frac{\varepsilon(t-1)}{2A} - \frac{4u}{3} + \frac{\varepsilon(t-1)}{A} \leq m^t(\pi) \leq \frac{\varepsilon(t-1)}{2A} + \frac{4u}{3} + \frac{\varepsilon(t-1)}{A}$

$\frac{\varepsilon(t-1)}{2A} - \frac{4u}{3}$

$$m^t(\pi) \geq \frac{\varepsilon(t-1)}{2A} - \frac{4u}{3}. \tag{2.12}$$

Setting $u = \log(2AT/\delta)$ and taking a union bound, we are guaranteed that with probability at least $1 - \delta$, for all $\pi \in \Pi$ and $t \in [T]$ (any fixed π에서 for all $\pi \in \Pi$로)

$$m^t(\pi) \geq \frac{\varepsilon(t-1)}{2A} - \frac{4\log(2AT/\delta)}{3}. \tag{2.13}$$

As long as $\varepsilon t \gtrsim A\log(AT/\delta)$ (we can write off the rounds where this does not hold), this yields

$$n^t(\pi) \geq m^t(\pi) \gtrsim \frac{\varepsilon t}{A}.$$

([illegible] shown that $n^t(\pi)$ is sufficiently large.)

Taking a union bound and combining with (2.11), this implies that with probability at least $1 - \delta$, for all t,

$$\max_\pi |f^\star(\pi) - \widehat{f}^t(\pi)| \lesssim \sqrt{\frac{A\log(AT/\delta)}{\varepsilon t}}.$$

proof of 2.9 ■

which leads to the overall regret bound

$$\textbf{Reg} \leq \sum_{t=1}^{T} \max_\pi |f^\star(\pi) - \widehat{f}^t(\pi)| + \varepsilon T \lesssim \sum_{t=1}^{T} \sqrt{\frac{A\log(AT/\delta)}{\varepsilon t}} + \varepsilon T$$
$$\leq \sqrt{\frac{AT\log(AT/\delta)}{\varepsilon}} + \varepsilon T. \tag{2.14}$$

$u_t = 1, v_t = \sqrt{\frac{A\log(AT/\delta)}{\varepsilon t}}$

$(\sum u_t v_t)^2 \leq (\sum u_t^2)(\sum v_t^2)$

$\left(\sum \sqrt{\frac{A\log(AT/\delta)}{\varepsilon t}}\right)^2 \leq \sum \left(\frac{A\log(AT/\delta)}{\varepsilon t}\right)$

$\sum \sqrt{\frac{A\log(AT/\delta)}{\varepsilon t}} \leq \sqrt{\sum \left(\frac{A\log(AT/\delta)}{\varepsilon t}\right)}$

$\leq \sqrt{\left(\frac{AT\log(AT/\delta)}{\varepsilon t}\right)}$

$\left(\sum_{t=1}^{T} \frac{1}{\sqrt{t}} \leq \sqrt{T}\right)$

To balance the terms on the right-hand side, we set

$$\varepsilon \propto \left(\frac{A\log(AT/\delta)}{T}\right)^{1/3},$$

"$\mathbb{E}[\text{Reg}] \lesssim A^{\frac{1}{3}} T^{\frac{2}{3}} \log^{\frac{1}{3}}(AT/\delta)$"

which gives the final result. Proof of proposition 4 ■ □

This proof shows that the ε-Greedy strategy allows the learner to acquire information uniformly for all actions, but we pay for this in terms of regret (specifically, through the εT factor in the final regret bound (2.14)). This issue here is that the ε-Greedy strategy continually explores all actions, even though we might expect to rule out actions with very low reward after a relatively small amount of exploration. To address this shortcoming, we will consider more adaptive strategies.

Remark 9 (Explore-then-commit): A relative of ε-Greedy is the explore-then-commit (ETC) algorithm (e.g., Robbins [73], Langford and Zhang [57]), which uniformly explores actions for the first N rounds, then estimates rewards based on the data collected and commits to the greedy action for the remaining $T-N$ rounds. This strategy can be shown to attain $\mathbf{Reg} \lesssim A^{1/3}T^{2/3}$ for an appropriate choice of N, matching ε-Greedy.

친척-!

2.3 The Upper Confidence Bound (UCB) Algorithm

The next algorithm we will study for bandits is the Upper Confidence Bound (UCB) algorithm [56, 6, 13]. The UCB algorithm attains a regret bound of the order $\widetilde{O}(\sqrt{AT})$, which improves upon the regret bound for ε-Greedy, and is optimal (in a worst-case sense) up to logarithmic factors. In addition to optimality, the algorithm offers several secondary benefits, including adaptivity to favorable structure in the underlying reward function.

The UCB algorithm is based on the notion of *optimism in the face of uncertainty*, which is a general principle we will revisit throughout this text in increasingly rich settings. The idea behind the principle is that at each time t, we should adopt the most optimistic perspective of the world possible given the data collected so far, and then choose the decision π^t based on this perspective.

To apply the idea of optimism to the multi-armed bandit problem, suppose that for each step t, we can construct "confidence intervals"

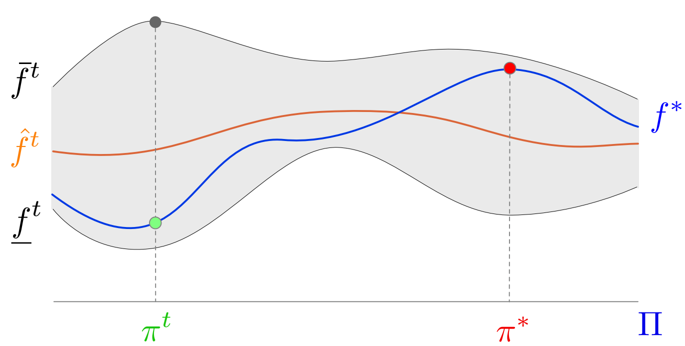
lower confi. bnd upper confi. bnd

$$\underline{f}^t, \bar{f}^t : \Pi \to \mathbb{R}, \tag{2.15}$$

with the following property: with probability at least $1-\delta$,

$$\forall t \in [T], \pi \in \Pi, \quad f^*(\pi) \in [\underline{f}^t(\pi), \bar{f}^t(\pi)]. \tag{2.16}$$

We refer to $\underline{f}^t$ as a *lower confidence bound* and $\bar{f}^t$ as a *upper confidence bound*, since we are



the instantaneous regret is bounded by "width" of confi. interval.

Figure 5: Illustration of the UCB algorithm. Selecting the action π^t optimistically ensures that the suboptimality never greater exceeds the confidence width.

guaranteed that with high probability, they lower (resp. upper) bound $f^\star$. Given confidence intervals, the UCB algorithm simply chooses π^t as the "optimistic" action that maximizes the upper confidence bound:

$$\pi^t = \arg\max_{\pi \in \Pi} \bar{f}^t(\pi).$$

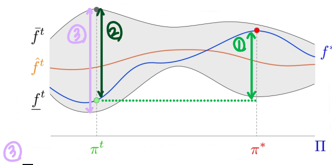
upper confi. bnd를 "최대화"

The following lemma shows that the instantaneous regret for this strategy is bounded by the width of the confidence interval; see Figure 5 for an illustration.

* UCB: chooses π^t ("optimistic action") that maximizes the upper bnd. $\pi^t = \arg\max_{\pi \in \Pi} \bar{f}^t(\pi)$.

Lemma 7: Fix t, and suppose that $f^\star(\pi) \in [\underline{f}^t(\pi), \bar{f}^t(\pi)]$ for all π. Then the optimistic action

$$\pi^t = \arg\max_{\pi \in \Pi} \bar{f}^t(\pi)$$



has

$$\overset{1}{f^\star(\pi^\star) - f^\star(\pi^t)} \leq \overset{2}{\bar{f}^t(\pi^t) - f^\star(\pi^t)} \leq \overset{3}{\bar{f}^t(\pi^t) - \underline{f}^t(\pi^t)}. \tag{2.17}$$

Proof of Lemma 7. The result follows immediate from the observation that for any $t \in [T]$ and any $\pi^\star \in \Pi$, we have

$f^\star(\pi^t) \geq \underline{f}^t(\pi^t)$

$$f^\star(\pi^\star) \leq \bar{f}^t(\pi^\star) \leq \bar{f}^t(\pi^t) \quad \text{and} \quad -f^\star(\pi^t) \leq -\underline{f}^t(\pi^t).$$

by def, ↑ π^t "maximizes" the upper confidence bnd.

□

Lemma 7 implies that as long as we can build confidence intervals (for which the width $\bar{f}^t(\pi^t) - \underline{f}^t(\pi^t)$ shrinks) the regret for the UCB strategy will be small. To construct such intervals, here we appeal to Hoeffding's inequality for adaptive stopping times (Lemma 33).[7] As long as $r^t \in [0, 1]$, a union bound gives that with probability at least $1 - \delta$, for all $t \in [T]$ and $\pi \in \Pi$,

Sample mean, pop. mean

$$\underbrace{|\hat{f}^t(\pi) - f^\star(\pi)|}_{\text{차이가}} \leq \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}, \tag{2.18}$$

작다.

where we recall that $\hat{f}^t$ is the sample mean and $n^t(\pi) := \sum_{i<t} \mathbb{I}\{\pi^i = \pi\}$. This suggests that by choosing

$$\bar{f}^t(\pi) = \hat{f}^t(\pi) + \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}, \quad \text{and} \quad \underline{f}^t(\pi) = \hat{f}^t(\pi) - \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}, \tag{2.19}$$

we obtain a valid confidence interval. With this choice—along with Lemma 7—we are in a favorable position, because for a given round t, one of two things must happen:

- The optimistic action has high reward, so the instantaneous regret is small.
- The instantaneous regret is large, which by Lemma 7 implies that confidence width is large as well (and $n^t(\pi^t)$ is small). This can only happen a small number of times, since $n^t(\pi^t)$ will increase as a result, causing the width to shrink.

Using this idea, we can prove the following regret bound.

Proposition 5: Using the confidence bounds in (2.19), the UCB algorithm ensures that with probability at least $1 - \delta$,

$$\mathbf{Reg} \lesssim \sqrt{AT\log(AT/\delta)}.$$

$\bar{f}^t = \hat{f}^t(\pi) + \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}$

$\underline{f}^t = \hat{f}^t(\pi) - \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}$

$\Pr\left(\mathbf{Reg} \gtrsim \sqrt{AT\log(AT/\delta)}\right) \leq \delta$

[7] While asymptotic confidence intervals in classical statistics arise from limit theorems, we are interested in valid *non-asymptotic* intervals, and thus appeal to concentration inequalities.

$\bar{f}^t = \hat{f}^t(\pi) + \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}$

$\underline{f}^t = \hat{f}^t(\pi) - \sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi)}}$

$\Pr\left(\mathrm{Reg} \gtrsim \sqrt{AT\log(AT/\delta)}\right) \le \delta$

This result is optimal up to the $\log(AT)$ factor, which can be removed by using the same algorithm with a slightly more sophisticated confidence interval construction [11]. Note that compared to the statistical learning and online learning setting, where we were able to attain regret bounds that scaled logarithmically with the size of the benchmark class, here the optimal regret scales *linearly* (not logarithmically) with $|\Pi| = A$. This is the price we pay for partial/bandit feedback, and reflects that fact that we must explore all actions to learn.

Proof of Proposition 5. Let us condition on the event in (2.18). Whenever this occurs, we have that $f^\star(\pi) \in [\underline{f}^t(\pi), \bar{f}^t(\pi)]$ for all $t \in [T]$ and $\pi \in \Pi$, so the confidence intervals are valid. As a result, Lemma 7 bounds regret in terms of the confidence width:

Step 1: bound the regret by the sum of confidence widths

$$\underbrace{\sum_{t=1}^{T} f^\star(\pi^\star) - f^\star(\pi^t)}_{\text{"Regret"}} \overset{\text{Lemma 7}}{\le} \underbrace{\sum_{t=1}^{T} \bar{f}^t(\pi^t) - \underline{f}^t(\pi^t)}_{\text{Regret... Bounded by the width}} \overset{\text{(2.18) but no abs. sign}}{\le} \sum_{t=1}^{T} 2\sqrt{\frac{2\log(2T^2A/\delta)}{n^t(\pi^t)}} \wedge 1; \tag{2.20}$$

there... logical and?

here, the "$\wedge 1$" term appears because we can write off the regret for early rounds where $n^t(\pi^t) = 0$ as 1.

To bound the right-hand side, we use a potential argument. The basic idea is that at every round, $n^t(\pi)$ must increase for some action π, and since there are only A actions, this means that $1/\sqrt{n^t(\pi^t)}$ can only be large for a small number of rounds. This can be thought of as a quantitative instance of the pigeonhole principle.

Lemma 8 (Confidence width potential lemma): We have

Step 2: Show that the sum of confidence widths is small.

$$\sum_{t=1}^{T} \frac{1}{\sqrt{n^t(\pi^t)}} \wedge 1 \lesssim \sqrt{AT}.$$

for각 Round

time t를 따라가면서 policy π가 선택된 시점들을 $\tau_1(\pi) < \tau_2(\pi) < \cdots < \tau_k(\pi) < \cdots < \tau_{n^{T+1}(\pi)}(\pi)$라 해보자.

이 시점 직전까지, $n^{\tau_1(\pi)} = 0$

이 시점 직전까지, $n^{\tau_2(\pi)} = 1$

이 시점 직전까지, $n^{\tau_k(\pi)} = k-1$.

↳ $\sum_{t=1}^{T} \frac{\mathbb{1}\{\pi^t = \pi\}}{\sqrt{n^t(\pi)}} \wedge 1 = \sum_{k=1}^{n^{T+1}(\pi)} \frac{1}{\sqrt{k-1}} \wedge 1.$

re-indexing

Proof of Lemma 8. We begin by writing.

$$\sum_{t=1}^{T} \frac{1}{\sqrt{n^t(\pi^t)}} \wedge 1 = \sum_{\pi} \sum_{t=1}^{T} \frac{\mathbb{I}\{\pi^t = \pi\}}{\sqrt{n^t(\pi)}} \wedge 1 = \sum_{\pi} \sum_{t=1}^{n^{T+1}(\pi)} \frac{1}{\sqrt{t-1}} \wedge 1. \tag{2.21}$$

↳시점마다...

For any $n \in \mathbb{N}$, we have $\sum_{t=1}^{n} \frac{1}{\sqrt{t-1}} \wedge 1 \le 1 + 2\sqrt{n}$, which allows us to bound (the whole LHS) by

$\cdots \left(\sum_{t=1}^{T} \frac{1}{\sqrt{t}} \le \sqrt{T}\right)$

or.. ① $k=1$: $1/\sqrt{0} \wedge 1 = 1$. ② $k=2$ to m: $\sum_{k=2}^{m} \frac{1}{\sqrt{k-1}} \wedge 1$. $\to \sum_{k=1}^{m} \frac{1}{\sqrt{k-1}} \wedge 1 \le 1 + 2\sqrt{m}$

$\sum_{k=2}^{m} \frac{1}{\sqrt{k-1}} \le \int_1^m \frac{1}{\sqrt{x}}\,dx = 2\sqrt{m} - 2 \le 2\sqrt{m}$

$$\underbrace{A}_{\substack{\text{size of the policy space }\Pi\ (|\Pi|=A) \\ \text{lower order term}}} + 2\sum_{\pi} \underbrace{\sqrt{n^T(\pi)}}_{\sqrt{\text{sigma (sum)}}}.$$

The factor of A above is a lower-order term (recall that we have $A \le \sqrt{AT}$ whenever $A \le T$, and if $A > T$ the regret bound we are proving is vacuous). To bound the second term, using Jensen's inequality, we have

$$\sum_{\pi} \sqrt{n^T(\pi)} \overset{\text{Jensen's Ineq.}}{\le} A\sqrt{\sum_{\pi} \frac{n^T(\pi)}{A}} = A\sqrt{T/A} = \sqrt{AT}.$$

↑ Just total number of rounds.

□

The main regret bound now follows from Lemma 8 and (2.20).

↳ $\Pr\left(\mathrm{Reg} \gtrsim \sqrt{AT\log(AT/\delta)}\right) \le \delta$

lemma8 (2.20)

□

To summarize, the key steps in the proof of Proposition 5 were to:

Step 1

1. Use the optimistic property and validity of the confidence bounds to bound regret by the sum of confidence widths.

Step 2

2. Use a potential argument to show that the sum of confidence widths is small.

We will revisit and generalize both ideas in subsequent chapters for more sophisticated settings, including contextual bandits, structured bandits, and reinforcement learning.

$O(\sqrt{AT \log(AT/\delta)})$

최악의 경우에..., (즉 어떤 경우이라도) 최선에 도달하는 알고리즘. (regret가 무조건 scale at $\Omega(\sqrt{AT})$ 하는 모델 M^*이 무조건 존재)

Remark 10 (Instance-dependent regret for UCB): The $\widetilde{O}(\sqrt{AT})$ regret bound attained by UCB holds uniformly for all models, and is (nearly) minimax-optimal, in the sense that for any algorithm, there exists a model $M^\star$ for which the regret must scale as $\Omega(\sqrt{AT})$. Minimax optimality is a useful notion of performance, but may be overly pessimistic. As an alternative, it is possible to show that the UCB attains what is known as an *instance-dependent* regret bound, which adapts to the underlying reward function, and can be smaller for "nice" problem instances.

Ωn 최악의 경우만 가정하냐~

Let $\Delta(\pi) := f^\star(\pi^\star) - f^\star(\pi)$ be the *suboptimality gap* for decision π. Then, when $f^\star(\pi) \in [0, 1]$, UCB can be shown to achieve

$$\mathbf{Reg} \lesssim \sum_{\pi:\Delta(\pi)>0} \frac{\log(AT/\delta)}{\Delta(\pi)}.$$

If we keep the underlying model fixed and take $T \to \infty$, this regret bound scales only *logarithmically* in T, which improves upon the $\sqrt{T}$-scaling of the minimax regret bound.

Not linearly!

2.4 Bayesian Bandits and the Posterior Sampling Algorithm*

underlying model... drawn from a known prior!

Up to this point, we have been designing and analyzing algorithms from a *frequentist* viewpoint, in which we aim to minimize regret for a *worst-case* choice of the underlying model $M^\star$. An alterative is to adopt a *Bayesian* viewpoint, and assume that the underlying model is drawn from a known *prior* $\mu \in \Delta(\mathcal{M})$.[8] In this case, rather than worst-case performance, we will be concerned with average regret under the prior, defined via

최악의 경우에~

(may) Not the worst choice case! (be)

$$\mathbf{Reg}_{\mathsf{Bayes}}(\mu) := \mathbb{E}_{M^\star\sim\mu}\, \mathbb{E}^{M^\star}[\mathbf{Reg}],$$

where $\mathbb{E}^{M^\star}[\cdot]$ denotes the algorithm's expected regret when $M^\star$ is the underlying reward distribution.

Working in the Bayesian setting opens up additional avenues for designing algorithms, because we can take advantage of our knowledge of the prior to compute quantities of interest that are not available in the frequentist setting, such as posterior distribution over $\pi^\star$ (after observing the dataset $\mathcal{H}^{t-1}$.) The most basic and well-known strategy here is *posterior sampling* (also known as Thompson sampling or probability matching) [82, 75].

[8]It is important that μ is known, otherwise this is no different from the frequentist setting.

Posterior Sampling

for $t = 1, \ldots, T$ **do**

Set $p^t(\pi) = \overbrace{\mathbb{P}(\pi^\star = \pi \mid \mathcal{H}^{t-1})}^{\text{posterior dist.}}$, where $\mathcal{H}^{t-1} = (\pi^1, r^1), \ldots, (\pi^{t-1}, r^{t-1})$.

Sample $\pi^t \sim p^t$ and observe r^t.

learner's action from posterior dist. of $\pi^\star$.

The basic idea is as follows. At each time t, we can use our knowledge of the prior to compute the distribution $\mathbb{P}(\pi^\star = \cdot \mid \mathcal{H}^{t-1})$, which represents the posterior distribution over $\pi^\star$ given all of the data we have collected from rounds $1, \ldots, t-1$. The posterior sampling algorithm simply samples the learner's action π^t from this distribution, thereby "matching" the posterior distribution of $\pi^\star$.

(this posterior dist) ("probability matching")

Proposition 6: For any prior μ, the posterior sampling algorithm ensures that

$$\mathbf{Reg}_{\mathsf{Bayes}}(\mu) \le \sqrt{AT \log(A)}. \tag{2.22}$$

In what follows, we prove a simplified version of Proposition 6; the full proof is given in Section 2.6.

Proof of Proposition 6 (simplified version). We will make the following simplified assumptions:

- We restrict to reward distributions where $\overset{\text{reward dist.}}{M^\star(\cdot \mid \pi)} = \mathcal{N}(f^\star(\pi), 1)$. That is, $f^\star$ is the only part of the reward distribution that is unknown.
  the mean reward function is the only part of the reward dist $M^\star$ that is "unknown"
  (Variance 고정, mean만 모름.)
- $f^\star$ belongs to a known class $\mathcal{F}$, and rather than proving the regret bound in Proposition 6, we will prove a bound of the form

  $$\mathbf{Reg}_{\mathsf{Bayes}}(\mu) \lesssim \sqrt{AT \log|\mathcal{F}|},$$

  which replaces the $\log A$ factor in the proposition with $\log|\mathcal{F}|$.

Since the mean reward function $f^\star$ is the only part of the reward distribution $M^\star$ that is unknown, we can simplify by considering an equivalent formulation where the prior has the form $\mu \in \Delta(\mathcal{F})$. That is, we have a prior over $f^\star$ rather than $M^\star$.

Before proceeding, let us introduce some notation. The process through which we sample $f^\star \sim \mu$ and the run the bandit algorithm induces a joint law over $(f^\star, \mathcal{H}^T)$, which we call $\mathbb{P}$. Throughout the proof, we use $\mathbb{E}[\cdot]$ to denote the expectation under this law. We also define $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{H}^t]$ and $\mathbb{P}_t[\cdot] = \mathbb{P}[\cdot \mid \mathcal{H}^t]$.

We begin by using the law of total expectation to express the expected regret as

↳ $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$

thee... expected regret.

$$\mathbf{Reg}_{\mathsf{Bayes}}(\mu) = \mathbb{E}\left[\sum_{t=1}^T \mathbb{E}_{t-1}[f^\star(\pi_{f^\star}) - f^\star(\pi^t)]\right].$$

Above, we have written $\pi^\star = \pi_{f^\star}$ to make explicit the fact that this is a random variable whose value is a function of $f^\star$.

We first simplify the expected regret for each step t. Let $\overset{\text{posterior dist.}}{\mu^t(f)} := \mathbb{P}(f^\star = f \mid \mathcal{H}^{t-1})$ be the posterior distribution at timestep t. The learner's decision π^t is *conditionally independent* of $f^\star$ given $\mathcal{H}^{t-1}$, so we can write

$$\mathbb{E}_{t-1}[f^\star(\pi_{f^\star}) - f^\star(\pi^t)] = \mathbb{E}_{f^\star \sim \mu^t, \pi^t \sim p^t}[f^\star(\pi_{f^\star}) - f^\star(\pi^t)].$$

RV (a function of f^*)

$\mathrm{Reg}_{\mathrm{Bayes}}(\mu) = \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{t-1}\left[f^*(\pi_{f^*}) - f^*(\pi^t)\right]\right]$ (LTE)

$\mathbb{E}_{f^*\sim\mu^t, \pi^t\sim p^t}\left[f^*(\pi_{f^*}) - f^*(\pi^t)\right]$ (π^t is condi. indep of f^* given $\mathcal{H}^{t-1}$)

If we define $\bar{f}^t(\pi) = \mathbb{E}_{f^\star\sim\mu^t}[f^\star(\pi)]$ as the expected reward function under the μ^t posterior, we can further write this as

$$\mathbb{E}_{f^\star\sim\mu^t, \pi^t\sim p^t}\left[f^\star(\pi_{f^\star}) - \bar{f}^t(\pi^t)\right].$$

By the design of the posterior sampling algorithm, $\pi^t \sim p^t$ is identical in distribution to $\pi_{f^\star}$ under $f^\star \sim \mu^t$, so this is equal to

$$\mathbb{E}_{f^\star\sim\mu^t}\left[f^\star(\pi_{f^\star}) - \bar{f}^t(\pi_{f^\star})\right].$$

This quantity captures—on average—how far a given realization of $f^\star$ deviates from the posterior mean $\bar{f}^t$, for a specific decision $\pi_{f^\star}$ (which is coupled to $f^\star$). The expression above might appear to be unrelated to the learner's decision distribution, but the next lemma shows that it is possible to relate this quantity back to the learner's decision distribution using a notion of *information gain* (or, estimation error).

Lemma 9: $\mathbb{E}_{f^*\sim\mu^t}\left[f^*(\pi_{f^*}) - \bar{f}(\pi_{f^*})\right] \le \sqrt{A \cdot \mathbb{E}_{f^*\sim\mu^t}\mathbb{E}_{\pi^t\sim p^t}\left[(f^*(\pi^t) - \bar{f}(\pi^t))^2\right]}$

Lemma 9 (Decoupling): For any function $\bar{f} : \Pi \to \mathbb{R}$, it holds that

$$\mathbb{E}_{f^\star\sim\mu^t}\left[f^\star(\pi_{f^\star}) - \bar{f}(\pi_{f^\star})\right] \le \sqrt{A \cdot \mathbb{E}_{f^\star\sim\mu^t}\,\mathbb{E}_{\pi^t\sim p^t}\left[(f^\star(\pi^t) - \bar{f}(\pi^t))^2\right]}. \tag{2.23}$$

Proof of Lemma 9. We will show a more general result. Namely, for any $\nu \in \Delta(\mathcal{F})$ and $\bar{f} : \Pi \to \mathbb{R}$, if we define $p(\pi) = \mathbb{P}_{f\sim\nu}(\pi_f = \pi)$, then

$$\mathbb{E}_{f\sim\nu}\left[f(\pi_f) - \bar{f}(\pi_f)\right] \le \sqrt{A \cdot \mathbb{E}_{f\sim\nu}\,\mathbb{E}_{\pi\sim p}\left[(f(\pi) - \bar{f}(\pi))^2\right]}. \tag{2.24}$$

This can be thought of as a "decoupling" lemma. On the left-hand side, the random variables f and π_f are coupled, but on the right-hand side, π is drawn from the *marginal distribution* over π_f, independent of the draw of f itself.

To prove the result, we use Cauchy-Schwarz as follows:

$$\begin{aligned}\mathbb{E}_{f\sim\nu}\left[f(\pi_f) - \bar{f}(\pi_f)\right] &= \mathbb{E}_{f\sim\nu}\left[\frac{p^{1/2}(\pi_f)}{p^{1/2}(\pi_f)}\big(f(\pi_f) - \bar{f}(\pi_f)\big)\right] \\ &\overset{\text{Cauchy Schwarz}}{\le} \left(\mathbb{E}_{f\sim\nu}\left[\frac{1}{p(\pi_f)}\right]\right)^{1/2} \cdot \left(\mathbb{E}_{f\sim\nu}\left[p(\pi_f)\big(f(\pi_f) - \bar{f}(\pi_f)\big)^2\right]\right)^{1/2}.\end{aligned}$$

For the first term, we have
(...의 루트 안에 있는거)

$\sum_{f:\pi_f=\pi} \nu(f) = p(\pi)$

$$\mathbb{E}_{f\sim\nu}\left[\frac{1}{p(\pi_f)}\right] \overset{\text{def. expectation}}{=} \sum_{f}\frac{\nu(f)}{p(\pi_f)} = \sum_{\pi}\sum_{f:\pi_f=\pi}\frac{\nu(f)}{p(\pi)} = \sum_{\pi}\frac{p(\pi)}{p(\pi)} = A.$$

For the second term, we have
(...의 루트 안에 있는거)

$$\mathbb{E}_{f\sim\nu}\left[p(\pi_f)\big(f(\pi_f) - \bar{f}(\pi_f)\big)^2\right] \le \mathbb{E}_{f\sim\nu}\left[\sum_{\pi} p(\pi)\big(f(\pi) - \bar{f}(\pi)\big)^2\right] = \mathbb{E}_{f\sim\nu}\,\mathbb{E}_{\pi\sim p}\left[(f(\pi) - \bar{f}(\pi))^2\right].$$

Putting these bounds together yields (2.24). □

Lemma 9:
$\mathbb{E}_{f^* \sim \mu^t}[f^*(\pi_{f^*}) - \bar{f}(\pi_{f^*})] \leq \sqrt{A \cdot \mathbb{E}_{f^* \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}[f^*(\pi^t) - \bar{f}(\pi^t))^2]}$

Using Lemma 9, we have that

add "$\mathbb{E} \sum_{t=1}^{T}$" then its $\mathrm{Reg}_{\mathrm{Bayes}}(\mu)$ (the expected regret).

$$\mathbb{E}[\mathbf{Reg}] \overset{\text{Lemma 9}}{\leq} \mathbb{E}\left[\sum_{t=1}^{T} \sqrt{A \cdot \mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2\right]}\right]$$
$$\leq \sqrt{AT \cdot \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2\right]\right]}.$$

k: t terms...
$\mathbb{E}[\sum^{T} \sqrt{AK}]$ vs. $\leq$ $\sqrt{AT\,\mathbb{E}[\sum^{T} K]}$
$= \sqrt{A}\,\mathbb{E}[\sum^{T} \sqrt{K}]$
$\mathbb{E}[\sum^{T} \sqrt{K}]$ vs. $\leq$ $\sqrt{T}\sqrt{\mathbb{E}[\sum^{T} K]}$

To finish up we will show that $\sum_{t=1}^{T} \mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2\right] \leq \log|\mathcal{F}|$. To do this, we need some additional information-theoretic tools.

그냥 Some definitions...

definition of entropy

- For a random variable X with distribution $\mathbb{P}$, $\mathsf{Ent}(X) \equiv \mathsf{Ent}(\mathbb{P}) := \sum_x p(x) \log(1/p(x))$.
- For random variables X and Y, $\mathsf{Ent}(X \mid Y = y) := \mathsf{Ent}(\mathbb{P}_{X|Y=y})$ and $\mathsf{Ent}(X \mid Y) := \mathbb{E}_{y \sim p_Y}[\mathsf{Ent}(X \mid Y = y)]$.
- For distributions $\mathbb{P}$ and $\mathbb{Q}$, $D_{\mathsf{KL}}(\mathbb{P} \,\|\, \mathbb{Q}) = \sum_x p(x) \log(p(x)/q(x))$.

To keep notation as clear as possible going forward, let us use boldface script ($\boldsymbol{\pi}^t$, $\boldsymbol{\pi}^\star$, $\boldsymbol{f}^\star$, $\boldsymbol{\mathcal{H}}^t$) to refer to the abstract random variables under consideration, and use non-boldface script (π^t, $\pi^\star$, $f^\star$, $\mathcal{H}^t$) to refer to their realizations. Our aim will be to use the conditional entropy $\mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^t)$ as a potential function, and show that for each t,

$$\frac{1}{2}\mathbb{E}\left[\mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2\right]\right] = \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^{t-1}) - \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^t). \tag{2.25}$$

SUM

From here the result will follow, because

$$\frac{1}{2}\mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2\right]\right] = \sum_{t=1}^{T} \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^{t-1}) - \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^t)$$
$$\overset{\Sigma\text{ 연산시}}{=} \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^0) - \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^T)$$
$$\leq \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^0) \text{ oo.}$$
$$\leq \log|\mathcal{F}|, \text{ (} f^* \text{ belongs to the known class } F\text{)}$$

where the last inequality follows because the entropy of a random variable X over a set $\mathcal{X}$ is always bounded by $\log|\mathcal{X}|$.

We proceed to prove (2.25). To begin, we use Lemma 39, which implies that

$$\frac{1}{2}(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2 \leq D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{r}^t|f^\star,\pi^t,\mathcal{H}^{t-1}} \,\|\, \mathbb{P}_{\boldsymbol{r}^t|\pi^t,\mathcal{H}^{t-1}}\right).$$

and

$$\frac{1}{2}\mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[(f^\star(\pi^t) - \bar{f}^t(\pi^t))^2\right] = \mathbb{E}_{f^\star \sim \mu^t} \mathbb{E}_{\pi^t \sim p^t}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{r}^t|f^\star,\pi^t,\mathcal{H}^{t-1}} \,\|\, \mathbb{P}_{\boldsymbol{r}^t|\pi^t,\mathcal{H}^{t-1}}\right)\right]$$

Since KL divergence satisfies $\mathbb{E}_{x \sim \mathbb{P}_X}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{Y|X=x} \,\|\, \mathbb{P}_Y\right)\right] = \mathbb{E}_{y \sim \mathbb{P}_Y}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{X|Y=y} \,\|\, \mathbb{P}_X\right)\right]$, this is equal to

$$\mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{f}^\star|\pi^t,r^t,\mathcal{H}^{t-1}} \,\|\, \mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^{t-1}}\right)\right] = \mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^{t-1}}\right)\right]. \tag{2.26}$$

$D_{KL}(P_{Z|X=x,Y=y} \| P_{Z|X=x}) = \sum_z P_{Z|X=x,Y=y}(z|x,y) \log \frac{P_{Z|X=x,Y=y}(z|x,y)}{P_{Z|X=x}(z|x)}$

전체 expectation

$\mathbb{E}[D_{KL}(P_{Z|X=x,Y=y} \| P_{Z|X=x})] = \mathbb{E}_{x,y,z}\left[\log \frac{P_{Z|X=x,Y=y}(z|x,y)}{P_{Z|X=x}(z|x)}\right] = \mathbb{E}_x[H(Z|X=x)] - \mathbb{E}_{x,y}[H(Z|X=x,Y=y)] = H(Z|X) - H(Z|X,Y)$

(Entropy) $H(X) = -\sum p \log p = \mathbb{E}[-\log p]$

Taking the expectation of (2.26) over $\mathcal{H}^{t-1}$, we can write this as

$$\mathbb{E}\left[\mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^{t-1}}\right)\right]\right] = \mathbb{E}_{\mathcal{H}^{t-1}} \mathbb{E}_{\mathcal{H}^t|\mathcal{H}^{t-1}}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^{t-1}}\right)\right].$$

A simple exercise shows that for random variables X, Y, Z,

$$\mathbb{E}_{(x,y)\sim\mathbb{P}_{X,Y}}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{Z|X=x,Y=y} \,\|\, \mathbb{P}_{Z|X=x}\right)\right] = \mathsf{Ent}(Z \mid X) - \mathsf{Ent}(Z \mid X, Y).$$

Applying this result above (and using that $\mathcal{H}^{t-1} \subset \mathcal{H}^t$) gives

$$\mathbb{E}_{\mathcal{H}^{t-1}} \mathbb{E}_{\mathcal{H}^t|\mathcal{H}^{t-1}}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{f}^\star|\mathcal{H}^{t-1}}\right)\right] = \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^{t-1}) - \mathsf{Ent}(\boldsymbol{f}^\star \mid \boldsymbol{\mathcal{H}}^t)$$

which was...

$\mathbb{E}\left[\frac{1}{2}\mathbb{E}_{f^\star\sim\mu^t}\mathbb{E}_{\pi^t\sim p^t}\left[(f^{M}(\pi^t) - \bar{f}^t(\pi^t))^2\right]\right]$.

as desired. □

The analysis above critically makes use of the fact that we are concerned with Bayesian regret, and have access to the *true prior*. One might hope that by choosing a sufficiently uninformative prior는, this approach might continue to work in the frequentist setting. In fact, this indeed is the case for bandits, though a different analysis is required [7, 8]. However, one can show (Sections 4 and 6) that the Bayesian analysis we have given here extends to significantly richer decision making settings, while the frequentist counterpart is limited to simple variants of the multi-armed bandit.

Remark 11 (Equivalence of min-max frequentist regret and max-min Bayesian regret): Using the minimax theorem, it is possible to show that under appropriate technical conditions

$$\min_{\mathsf{Alg}} \max_{M^\star} \overbrace{\mathbb{E}^{M^\star,\mathsf{Alg}}[\mathbf{Reg}]}^{\text{frequentist Regret}} = \max_{\mu\in\Delta(\mathcal{M})} \min_{\mathsf{Alg}} \overbrace{\mathbb{E}_{M^\star\sim\mu}\,\mathbb{E}^{M^\star,\mathsf{Alg}}[\mathbf{Reg}]}^{\text{Bayesian Regret}}.$$

worst-case의 최선. 최선의 worst-case. RHS: worst-case value of the Bayesian regret.

That is, if we take the worst-case value of the Bayesian regret over all possible choices of prior, this coincides with the minimax value of the frequentist regret.

2.5 Adversarial Bandits and the Exp3 Algorithm*

We conclude this section with a brief introduction to the multi-armed bandit problem with non-stochastic/adversarial rewards, which dispenses with Assumption 1. In the context of Figure 2, the non-stochastic nature of rewards adds a new "adversarial data" dimension to the problem. As one might expect, the solution we will present for non-stochastic bandits will leverage the the online learning tools introduced in Section 1.6.

To simplify the presentation, suppose that the collection of rewards

$$\{r^t(\pi) \in [0,1] : \pi \in [A], t \in [T]\}$$

for each action and time step is arbitrary and fixed ahead of the interaction by an oblivious adversary. Since we do not posit a stochastic model for rewards, we define regret as in (2.4).

The algorithm we present will build upon the exponential weights algorithm studied in the context of online supervised learning in Section 1.6. To make the connection as clear as possible, we make a temporary switch from rewards to losses, mapping r^t to $1 - r^t$, a transformation that does not change the problem itself.

Recall that p^t denotes the randomization distribution for the learner at round t. As discussed in Remark 7, we can write expected regret as

Expected Regret.

$$\mathbf{Reg} = \sum_{t=1}^{T} \langle p^t, \boldsymbol{\ell}^t \rangle - \min_{\pi \in [A]} \sum_{t=1}^{T} \langle \boldsymbol{e}_\pi, \boldsymbol{\ell}^t \rangle \tag{2.27}$$

where $\boldsymbol{\ell}^t \in [0,1]^A$ is the vector of losses for each of the actions at time t.

Since only the loss (equivalently, reward) of the chosen action $\pi^t \sim p^t$ is observed, we cannot directly appeal to the exponential weights algorithm, which requires knowledge of the full vector $\boldsymbol{\ell}^t$. To address this, we build an *unbiased estimate* of the vector $\boldsymbol{\ell}^t$ from a single real-valued observation $\boldsymbol{\ell}^t(\pi^t)$. At first, this might appear impossible, but it is straightforward to show that

$$\mathbb{E}_{\pi^t\sim p^t}\left[\hat{\ell}^t(\pi|\pi^t)\right] = \sum_{\pi' \in [A]} p^t(\pi) \tilde{\ell}^t(\pi|\pi')$$

an "unbiased estimate" $\tilde{\ell}^t(\pi)$

$$\widetilde{\boldsymbol{\ell}}^t(\pi) = \frac{\boldsymbol{\ell}^t(\pi)}{p^t(\pi)} \times \mathbb{I}\{\pi^t = \pi\} \tag{2.28}$$

$$= \sum_{\pi' \in [A]} p^t(\pi') \frac{\ell^t(\pi)}{p^t(\pi)} \mathbb{1}\{\pi^t = \pi\}$$

is an unbiased estimate for all $\pi \in [A]$, or in vector notation

$$= p^t(\pi) \frac{\ell^t(\pi)}{p^t(\pi)} = \ell^t(\pi)$$

$\therefore$ unbiased

$$\mathbb{E}_{\pi^t \sim p^t}\left[\widetilde{\boldsymbol{\ell}}^t\right] = \boldsymbol{\ell}^t. \tag{2.29}$$

If we apply the exponential weights algorithm with the loss vectors $\widetilde{\boldsymbol{\ell}}^t$, it can be shown to attain regret

이 과정: exercise.

결론: $\mathbb{E}[Reg] \lesssim \sqrt{AT \log A}$

$$\mathbb{E}_{\pi^t\sim p^t}\left[\langle p^t, \tilde{\ell}^t\rangle\right] = \mathbb{E}_{\pi^t\sim p^t}\left[\sum_\pi p^t(\pi)\hat{\ell}_t(\pi|\pi^t)\right] = \sum_\pi p^t(\pi)\mathbb{E}_{\pi^t\sim p^t}\left[\hat{\ell}^t(\pi|\pi^t)\right] = \sum_\pi p^t(\pi)\ell^t(\pi) = \langle p^t, \ell^t\rangle.$$

$$\langle e_\pi, \hat{\ell}^t\rangle = \sum_{j=1}^{A} e_\pi(j)\tilde{\ell}^t(j|\pi^t) = \overset{1}{e_\pi(\pi)}\tilde{\ell}^t(\pi|\pi^t) + \sum_{j\neq\pi} \underbrace{e_\pi(j)}_{0}\tilde{\ell}^t(j|\pi^t) = \tilde{\ell}^t(\pi|\pi^t)$$

$$\mathbb{E}[\mathbf{Reg}] = \mathbb{E}\left[\sum_{t=1}^{T} \langle p^t, \boldsymbol{\ell}^t \rangle\right] - \min_{\pi} \sum_{t=1}^{T} \langle e_\pi, \boldsymbol{\ell}^t \rangle \tag{2.30}$$

oo.

마찬가지로 $\mathbb{E}_{\pi^t\sim p^t}\left[\langle e_\pi, \tilde{\ell}^t\rangle\right] = \langle e_\pi, \ell^t\rangle.$

$$= \mathbb{E}\left[\sum_{t=1}^{T} \left\langle p^t, \widetilde{\boldsymbol{\ell}}^t \right\rangle\right] - \min_{\pi} \mathbb{E}\left[\sum_{t=1}^{T} \left\langle e_\pi, \widetilde{\boldsymbol{\ell}}^t \right\rangle\right] \lesssim \sqrt{AT \log A}. \tag{2.31}$$

oo. subprob 1~3.

$$= \mathbb{E}\left[\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[\langle p^t, \tilde{\ell}^t\rangle\right]\right] - \min_\pi \mathbb{E}\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[\langle e_\pi, \tilde{\ell}^t\rangle\right]$$

$$\sum_{t=1}^{T}\langle e_\pi, \ell^t\rangle = \sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[\langle e_\pi, \tilde{\ell}^t\rangle\right]$$

This algorithm is known as *Exp3* ("Exponential Weights for Exploration and Exploitation"). A full proof of this result is left as an exercise in Section 2.7.

oo.

"min & expectation"

$$\min_\pi \sum_{t=1}^{T}\langle e_\pi, \ell^t\rangle = \min_\pi \mathbb{E}\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[\langle e_\pi, \tilde{\ell}^t\rangle\right]$$

2.6 Deferred Proofs

(In section 2.4)

Proof of Proposition 6 (full version) . Let $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{H}^t]$ and $\mathbb{P}_t[\cdot] = \mathbb{P}[\cdot \mid \mathcal{H}^t]$. We begin by using the law of total expectation to express the expected regret as

$$\mathbf{Reg}_{\mathsf{Bayes}}(\mu) = \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{t-1}[f^\star(\pi^\star) - f^\star(\pi^t)]\right].$$

Here and throughout the proof, $\mathbb{E}[\cdot]$ will denote the joint expectation over both $M^\star \sim \mu$ and over the sequence $\mathcal{H}^T = (\pi^1, r^1), \ldots, (\pi^T, r^T)$ that the algorithm generates by interacting with $M^\star$.

We first simplify the (conditional) expected regret for each step t. Let $\bar{f}^t(\pi) := \mathbb{E}_{t-1}[f^\star(\pi)]$ denote the *posterior mean reward* function at time t, which should be thought of as the expected value of $f^\star$ given everything we have learned so far. Next, let $\bar{f}^t_{\pi'}(\pi) = \mathbb{E}_{t-1}[f^\star(\pi) \mid \pi^\star = \pi']$, which is the expected reward given everything we have learned so far, assuming that $\pi^\star = \pi'$. We proceed to write the expression

$$\mathbb{E}_{t-1}[f^\star(\pi^\star) - f^\star(\pi^t)]$$

in terms of these quantities. For the learner's reward, we observe that $f^\star$ is conditionally independent of π^t given $\mathcal{H}^{t-1}$, we have

$$\mathbb{E}_{t-1}[f^\star(\pi^t)] = \mathbb{E}_{\pi\sim p^t}[\bar{f}^t(\pi)].$$

For the reward of the optimal action, we begin by writing

$$\begin{aligned}\mathbb{E}_{t-1}[f^\star(\pi^\star)] &= \sum_{\pi\in\Pi} \mathbb{P}_{t-1}(\pi^\star = \pi)\,\mathbb{E}_{t-1}[f^\star(\pi) \mid \pi^\star = \pi] \\ &= \sum_{\pi\in\Pi} \mathbb{P}_{t-1}(\pi^\star = \pi)\bar{f}^t_\pi(\pi) \\ &= \mathbb{E}_{\pi\sim p^t}\left[\bar{f}^t_\pi(\pi)\right],\end{aligned}$$

where we have used that p^t was chosen to match the posterior distribution over $\pi^\star$. This establishes that

$$\mathbb{E}_{t-1}[f^\star(\pi^\star) - f^\star(\pi^t)] = \mathbb{E}_{\pi\sim p^t}\left[\bar{f}^t_\pi(\pi) - \bar{f}^t(\pi)\right].$$

We now make use of the following decoupling-type inequality, which follows from (2.24):

$$\mathbb{E}_{\pi\sim p^t}\left[\bar{f}^t_\pi(\pi) - \bar{f}^t(\pi)\right] \le \sqrt{A \cdot \mathbb{E}_{\pi,\pi^\star\sim p^t}\left[(\bar{f}^t_{\pi^\star}(\pi) - \bar{f}^t(\pi))^2\right]}. \tag{2.32}$$

To keep notation as clear as possible going forward, let us use boldface script ($\boldsymbol{\pi}^t$, $\boldsymbol{\pi}^\star$, $\boldsymbol{f}^\star$, $\boldsymbol{\mathcal{H}}^t$) to refer to the abstract random variables under consideration, and use non-boldface script (π^t, $\pi^\star$, $f^\star$, $\mathcal{H}^t$) to refer to their realizations. As in the simplified proof, we will show that the right-hand side in (2.32) is related to a notion of *information gain* (that is, information about $\pi^\star$ acquired at step t). Using Pinsker's inequality, we have

$$\mathbb{E}_{\pi^t,\pi^\star\sim p^t}\left[(\bar{f}^t_{\pi^\star}(\pi^t) - \bar{f}^t(\pi^t))^2\right] \le \mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{r}^t|\pi^\star,\pi^t,\mathcal{H}^{t-1}} \,\|\, \mathbb{P}_{\boldsymbol{r}^t|\pi^t,\mathcal{H}^{t-1}}\right)\right].$$

Since KL divergence satisfies $\mathbb{E}_X\left[D_{\mathsf{KL}}\left(\mathbb{P}_{Y|X} \,\|\, \mathbb{P}_Y\right)\right] = \mathbb{E}_Y\left[D_{\mathsf{KL}}\left(\mathbb{P}_{X|Y} \,\|\, \mathbb{P}_X\right)\right]$, this is equal to

$$\mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{\pi}^\star|\pi^t,r^t,\mathcal{H}^{t-1}} \,\|\, \mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^{t-1}}\right)\right] = \mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^{t-1}}\right)\right]. \tag{2.33}$$

This is quantifying how much information about $\pi^\star$ we gain by playing π^t and observing r^t at step t, relative to what we knew at step $t-1$. Applying (2.32) and (2.33), we have

$$\begin{aligned}\sum_{t=1}^T \mathbb{E}_{\pi\sim p^t}\left[\bar{f}^t_\pi(\pi) - \bar{f}^t(\pi)\right] &\le \sum_{t=1}^T \sqrt{A \cdot \mathbb{E}_{\pi,\pi^\star\sim p^t}\left[(\bar{f}^t_{\pi^\star}(\pi) - \bar{f}^t(\pi))^2\right]} \\ &\le \sum_{t=1}^T \sqrt{A \cdot \mathbb{E}_{t-1}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^{t-1}}\right)\right]} \\ &\le \sqrt{AT \cdot \sum_{t=1}^T \mathbb{E}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^{t-1}}\right)\right]}.\end{aligned}$$

We can write

$$\mathbb{E}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^{t-1}}\right)\right] = \mathsf{Ent}(\boldsymbol{\pi}^\star \mid \boldsymbol{\mathcal{H}}^{t-1}) - \mathsf{Ent}(\boldsymbol{\pi}^\star \mid \boldsymbol{\mathcal{H}}^t),$$

so telescoping gives

$$\sum_{t=1}^T \mathbb{E}\left[D_{\mathsf{KL}}\left(\mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^t} \,\|\, \mathbb{P}_{\boldsymbol{\pi}^\star|\mathcal{H}^{t-1}}\right)\right] = \mathsf{Ent}(\boldsymbol{\pi}^\star \mid \boldsymbol{\mathcal{H}}^0) - \mathsf{Ent}(\boldsymbol{\pi}^\star \mid \boldsymbol{\mathcal{H}}^T) \le \log(A).$$

□

2.7 Exercises

Exercise 4 (Adversarial Bandits): In this exercise, we will prove a regret bound for *adversarial bandits* (Section 2.5), where the sequence of rewards (losses) is non-stochastic. To make a direct connection to the Exponential Weights Algorithm, we switch from rewards to losses, mapping r^t to $1 - r^t$, a transformation that does not change the problem itself. To simplify the presentation, suppose that a collection of losses

$$\{\boldsymbol{\ell}^t(\pi) \in [0,1] : \pi \in [A], t \in [T]\}$$

for each action π and time step t is arbitrary and chosen before round $t = 1$; this is referred to as an *oblivious adversary*. We denote by $\boldsymbol{\ell}^t = (\boldsymbol{\ell}^t(1), \ldots, \boldsymbol{\ell}^t(A))$ the vector of losses at time t. The protocol for the problem of adversarial multi-armed bandits (with losses) is as follows:

Multi-Armed Bandit Protocol
for $t = 1, \ldots, T$ **do**
Select decision $\pi^t \in \Pi := \{1, \ldots, A\}$ by sampling $\pi^t \sim p^t$
Observe loss $\boldsymbol{\ell}^t(\pi^t)$

Let p^t be the randomization distribution of the decision-maker on round t. Expected regret can be written as

$$\mathbb{E}[\mathbf{Reg}] = \mathbb{E}\left[\sum_{t=1}^{T} \langle p^t, \boldsymbol{\ell}^t \rangle\right] - \min_{\pi \in [A]} \sum_{t=1}^{T} \langle \boldsymbol{e}_\pi, \boldsymbol{\ell}^t \rangle. \tag{2.34}$$

Since only the loss of the chosen action $\pi^t \sim p^t$ is observed, we cannot directly appeal to the Exponential Weights Algorithm. The solution is to build an unbiased estimate of the vector $\boldsymbol{\ell}^t$ from the single real-valued observation $\boldsymbol{\ell}^t(\pi^t)$.

1. Prove that the vector $\widetilde{\boldsymbol{\ell}}^t(\cdot \mid \pi^t)$ defined by

$$\widetilde{\boldsymbol{\ell}}^t(\pi \mid \pi^t) = \frac{\boldsymbol{\ell}^t(\pi)}{p^t(\pi)} \times \mathbb{I}\{\pi^t = \pi\} \tag{2.35}$$

is an *unbiased estimate* for $\boldsymbol{\ell}^t(\pi)$ for all $\pi \in [A]$. In vector notation, this means

$$\mathbb{E}_{\pi^t \sim p^t}[\widetilde{\boldsymbol{\ell}}^t(\cdot \mid \pi^t)] = \boldsymbol{\ell}^t.$$

Conclude that

$$\mathbb{E}[\mathbf{Reg}] = \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} \langle p^t, \widetilde{\boldsymbol{\ell}}^t \rangle\right] - \min_{\pi \in [A]} \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t} \langle e_\pi, \widetilde{\boldsymbol{\ell}}^t \rangle\right] \tag{2.36}$$

Above, we use the shorthand $\widetilde{\boldsymbol{\ell}}^t = \widetilde{\boldsymbol{\ell}}(\cdot \mid \pi^t)$.

2. Show that given π',

$$\mathbb{E}_{\pi \sim p^t}\left[\widetilde{\boldsymbol{\ell}}^t(\pi \mid \pi')^2\right] = \frac{\boldsymbol{\ell}^t(\pi')^2}{p^t(\pi')}, \quad \text{so that} \quad \mathbb{E}_{\pi^t \sim p^t} \mathbb{E}_{\pi \sim p^t}\left[\widetilde{\boldsymbol{\ell}}^t(\pi \mid \pi^t)^2\right] \leq A. \tag{2.37}$$

3. Define

$$p^t(\pi) \propto \exp\left\{-\eta \sum_{s=1}^{t-1} \langle e_\pi, \widetilde{\boldsymbol{\ell}}^s(\cdot \mid \pi^s) \rangle\right\},$$

which corresponds to the exponential weights algorithm on the estimated losses $\widetilde{\boldsymbol{\ell}}^{s}$. Apply (1.41) to the estimated losses to show that for any $\pi \in [A]$,

$$\mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left\langle p^t, \widetilde{\boldsymbol{\ell}}^{t}\right\rangle\right] - \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left\langle e_\pi, \widetilde{\boldsymbol{\ell}}^{t}\right\rangle\right] \lesssim \sqrt{AT \log A}$$

Hence, the price of bandit feedback in the adversarial model, as compared to full-information online learning, is only $\sqrt{A}$.

3. CONTEXTUAL BANDITS

In the last section, we studied the multi-armed bandit problem, which arguably the simplest framework for interactive decision making. This simplicity comes at a cost: few real-world problems can be modeled as a multi-armed bandit problem directly. For example, for the problem of selecting medical treatments, the multi-armed bandit formulation presupposes that one treatment rule (action/decision) is good for all patients, which is clearly unreasonable. To address this, we augment the problem formulation by allowing the decision-maker to select the action π^t after observing a *context* x^t; this is called the *contextual bandit* problem. The context x^t, which may also be thought of as a feature vector or collection of

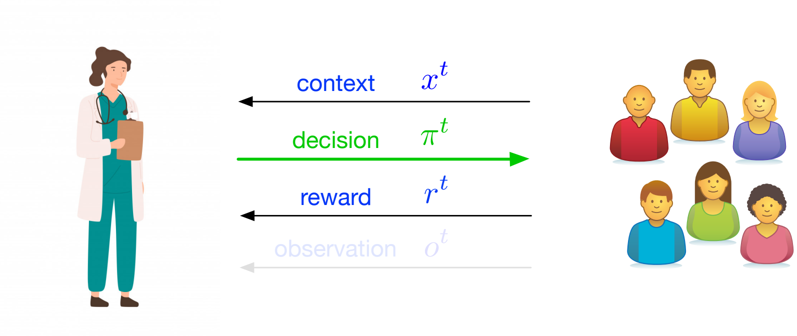


Figure 6: An illustration of the contextual multi-armed bandit problem. A doctor (the learner) aims to select a treatment based on the context (medical history, symptoms).

covariates (e.g., a patient's medical history, or the profile of a user arriving at a website), can be used by the learner to better maximize rewards by tailoring decisions to the specific patient or user under consideration.

Contextual Bandit Protocol
for $t = 1, \ldots, T$ **do**
Observe context $x^t \in \mathcal{X}$.
Select decision $\pi^t \in \Pi = \{1, \ldots, A\}$.
Observe reward $r^t \in \mathbb{R}$.

As with multi-armed bandits, contextual bandits can be studied in a stochastic framework or in an adversarial framework. In this course, we will allow the contexts $x^1, \ldots, x^T$ to be generated in an arbitrary, potentially adversarially fashion, but assume that rewards are generated from a fixed conditional distribution.